Events of the testbench

Event 1 — Reset (t = 0 to t = 8ns)

All 32 registers are set to 0. Both rd_data outputs show 0.

Event 2 — Single commit: x1 = 42 (posedge at t = 25ns)

commit_valid = 0001, commit_rd_0 = 1, commit_data_0 = 42 driven at t = 18ns. On posedge at t = 25ns, arf[1] = 42. Since rd_addr_0 = 1, rd_data_0 immediately becomes 42.

Event 3 — 4-wide commit (posedge at t = 35ns)

commit_valid = 1111 with x2=100, x3=200, x4=300, x5=400 driven at t = 28ns. On posedge at t = 35ns all four writes happen in the same cycle. rd_data_1 = arf[3] becomes 200.

Event 4 — Attempt to write x0 (posedge at t = 45ns)

commit_rd_0 = 0 and commit_data_0 = 999 driven at t = 38ns. On posedge at t = 45ns the condition commit_rd_0 != 0 fails so the write is blocked. arf[0] stays 0.

Event 5 — Switch read port to x0 (t = 53ns)

rd_addr_0 is switched to 0. Since the read is combinational and x0 is hardwired, rd_data_0 immediately shows 0 regardless of the attempted write.

Expected Signal Values

Time	Event	commit_valid	rd_data_0(arf[x1]→arf[x0])	rd_data_1 (arf[x3])
t = 0–8ns	Reset active	0000	0	0
t = 18ns	Step 1 driven: x1 = 42	0001	0	0
t = 25ns	POSEDGE — arf[1] = 42 latched	0001	42	0
t = 28ns	Step 2 driven: x2=100, x3=200, x4=300, x5=400	1111	42	0
t = 35ns	POSEDGE — arf[2..5] latched	1111	42	200
t = 38ns	Step 3 driven: x0 = 999 (attempt)	0001	42	200
t = 45ns	POSEDGE — blocked, x0 write ignored	0001	42	200
t = 48ns	commit_valid cleared	0000	42	200
t = 53ns	rd_addr_0 switched to x0	0000	0	200

Signals

Waves

Time	0	10 ns	20 ns	30 ns	40 ns	50 ns	60
clk							
rst							
commit_valid[3:0]	0000		0001	1111	0001	0000	
commit_data_0[63:0]	0		42	100	999		
commit_data_1[63:0]	0			200			
commit_data_2[63:0]	0			300			
commit_data_3[63:0]	0			400			
commit_rd_0[4:0]	0		1	2	0		
commit_rd_1[4:0]	0			3			
commit_rd_2[4:0]	0			4			
commit_rd_3[4:0]	0			5			
rd_addr_0[4:0]	1					0	
rd_addr_1[4:0]	3						
rd_data_0[63:0]	0			42		0	
rd_data_1[63:0]	0				200		